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Abstract

Joint time-vertex graph signal processing has seen significant advancements recently. This paper investigates the sampling and recovery in the joint time-vertex linear canonical transform (JLCT) domain. We prove that bandlimited signals in the JLCT domain can be perfectly recovered. Experimental design strategies are employed to generate optimal sampling operators on time-vertex graphs. Furthermore, numerical experiments demonstrate superior performance compared to methods based on the joint time-vertex Fourier transform and fractional Fourier transform.

CCS Concepts

• **Mathematics of computing** → *Spectra of graphs.*

Keywords

Graph signal processing, time-vertex signal processing, graph linear canonical transform, sampling.

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1 Introduction

With the continuous advancement of graph signal processing (GSP), its sampling theory has also matured, enabling the recovery of entire signals from partial samples, thereby significantly reducing storage and transmission costs [1–7]. However, most prior work has focused on static graph signals, whereas many real-world signals, such as temperature readings from sensor networks, are dynamic over time [8, 9]. These time-varying graph signals, represented as high-dimensional vectors or tensors at each vertex, have led to growing interest in joint time-vertex graph signal processing.

In recent years, sampling methods for joint time-vertex graph signals have garnered significant attention. The signal spectrum can be obtained via the joint time-vertex Fourier transform (JFT) [9]. For instance, R. Varma et al. defined smooth signals on joint time-vertex models using product graphs and proposed a recovery strategy [10]. K. Qiu et al. introduced a batch reconstruction method for time-varying graph signals based on the smoothness of temporal differences [11]. J. Yu et al. developed a critical sampling set algorithm with the minimum number of samples [12], while Ji et al. extended the time domain into Hilbert space, proposing a generalized graph signal sampling framework [13].

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Although JFT methods can sample time-vertex graph signals, they struggle to capture the transformation process from the time-vertex domain to the spectral domain and are limited in handling chirp-like features in graph signals. These methods also face challenges in addressing degrees of freedom, flexibility, parameter utilization, and effectively dealing with non-stationary signals and non-integer orders [15]. Inspired by sampling methods for non-stationary graph signals in static graphs and the joint time-vertex linear canonical transform (JLCT) introduced in [14], this work proposes a novel sampling framework based on the JLCT, which integrates the discrete linear canonical transform (DLCT) in the time domain with the graph linear canonical transform (GLCT) in the vertex domain.

The GLCT extends the graph Fourier transform (GFT) by introducing scaling, chirp modulation, and fractional-order operations, enabling it to model a broader class of signals, including those with non-linear spectral distributions and varying smoothness [14–16]. Building on this, the JLCT combines these capabilities into a unified framework for time-vertex analysis, capturing intricate temporal-graph correlations through its tensor-product structure [14]. This allows JLCT to generalize the concept of bandlimitedness, overcoming limitations of the JFT and joint time-vertex fractional Fourier transform (JFRFT) [17], and providing a more flexible basis for modeling non-stationary and multi-scale signals.

The key contribution of this work lies in designing optimal sampling operators tailored to the JLCT domain, leveraging experimental design strategies to maximize recovery accuracy and robustness against noise. By proving perfect recovery conditions for JLCT-bandlimited signals and demonstrating superior performance over JFT and JFRFT in both synthetic and real-world scenarios, this work not only extends existing sampling theories but also highlights the practical relevance of JLCT in applications such as dynamic sensor networks and time-varying geometric networks.

2 Background and Modeling

2.1 Graph Linear Canonical Transform and Its Sampling Theory

In GSP, the high-dimensional structure of the signal is represented by a graph $\mathcal{G} = (\mathcal{V}, \mathbf{A})$, where $\mathcal{V} = \{v_0, \ldots, v_{N-1}\}$ represents the set of vertices and $\mathbf{A} \in \mathbb{R}^{N\times N}$ represents the weighted adjacency matrix. The graph signal is defined as a mapping that maps vertices v_n to signal coefficients $x_n \in \mathbb{C}$. For a graph signal $\mathbf{x} \in \mathbb{C}^N$, the GFT [1] is defined by the eigendecomposition of an adjacency matrix $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$, where the columns of $\mathbf{V}$ are the eigenvectors of $\mathbf{A}$ and the corresponding eigenvalue diagonal matrix is $\mathbf{\Lambda}$, which is defined as $\hat{\mathbf{x}} = \mathbf{V}^{-1}\mathbf{x}$. And the operator is further eigendecomposed as

$\mathbf{V}^{-1} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{-1}$, yielding the eigenvector matrix $\mathbf{Q}$ and the diagonal eigenvalue matrix $\mathbf{\Lambda}$.

The introduction of the GLCT involves three stages: (i) when the operator is $\mathbf{Q}\mathbf{\Lambda}^{\alpha}\mathbf{Q}^{-1}$, it corresponds to the graph fractional Fourier transform (GFRFT); (ii) when the operator is $\mathbf{Q}_{\sigma}\mathbf{Q}^{-1}$, it represents graph scaling [15]; and (iii) when the operator is $\mathbf{\Lambda}^{\xi}$, it represents graph chirp multiplication. Integrating these stages, the GLCT is defined as [15]

$$\hat{\mathbf{x}} = (\mathbf{\Lambda}^{\xi})(\mathbf{Q}_{\sigma}\mathbf{Q}^{-1})(\mathbf{Q}\mathbf{\Lambda}^{\alpha}\mathbf{Q}^{-1})\mathbf{x} = (\mathbf{\Lambda}^{\xi}\mathbf{Q}_{\sigma}\mathbf{\Lambda}^{\alpha}\mathbf{Q}^{-1})\mathbf{x}. \tag{1}$$

For convenience, let $\mathbf{O}_{\mathcal{G}}^{\mathbf{M}} = \left(\mathbf{\Lambda}^{\xi}\mathbf{Q}_{\sigma}\mathbf{\Lambda}^{\alpha}\mathbf{Q}^{-1}\right) \in \mathbb{C}^{N\times N}$. The GLCT parameters (ξ, σ, α) are related to the GLCT parameter matrix $\mathbf{M} = [a, b; c, d]$, where $ad - bc = 1$ and $a, b, c, d \in \mathbb{R}$. Specifically, the relationships are $\xi = \frac{ac+bd}{a^2+b^2}, \sigma = \sqrt{a^2+b^2}, \alpha = \frac{2}{\pi}\arccos\left(\frac{a}{\sigma}\right) = \frac{2}{\pi}\arcsin\left(\frac{b}{\sigma}\right)$.

Sampling in the GLCT domain is similar to traditional GFT sampling [5], relying on bandlimited signals. Consider the sampling set $\mathcal{S} = (\mathcal{S}_0, \ldots, \mathcal{S}_i, \ldots, \mathcal{S}_{|\mathcal{S}|-1})$, where $\mathcal{S}_i \in \{0, 1, \ldots, N-1\}$, and the size of the set is $|\mathcal{S}|$. This size determines the indices used to sample the bandlimited graph signal $\mathbf{x} \in \mathbb{C}^N$, resulting in a sampled signal $\mathbf{x}_{\mathcal{S}} = \mathbf{D}\mathbf{x} \in \mathbb{C}^{|\mathcal{S}|}$ with $|\mathcal{S}| < N$. The sampling operator is defined as

$$\mathbf{D}(i, j) = \begin{cases} 1, & j \in \mathcal{S}_i, \\ 0, & \text{otherwise}, \end{cases} \tag{2}$$

where $\mathbf{D} \in \mathbb{C}^{|\mathcal{S}|\times N}$. We define a signal as $\mathbf{M}$-bandlimited [16] in the GLCT domain.

Definition 1. *The closed subspace of a graph signal* $\mathbf{x} \in \mathbb{C}^N$ *with bandwidth at most* $|\mathcal{F}|$ *is denoted as* $\mathbf{x} \in \mathrm{BL}_{|\mathcal{F}|}(\mathbf{O}_{\mathcal{G}}^{\mathbf{M}})$, *where* $\mathbf{x} = \mathbf{B}^{\mathbf{M}}\mathbf{x} = (\mathbf{O}_{\mathcal{G}}^{\mathbf{M}})^{-1}\Sigma_{\mathcal{F}}\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}\mathbf{x}$, *and* $\Sigma_{\mathcal{F}} = diag(d_1, \ldots, d_N)$, *where* $d_i = 1$ *if* $i \in \mathcal{F}$, *and* $d_i = 0$ *otherwise.*

If the sampling operator $\mathbf{D}$ satisfies $\mathrm{rank}\{\mathbf{D}(\mathbf{O}_{\mathcal{G}}^{\mathbf{M}})_{|\mathcal{F}|}^{-1}\} = |\mathcal{F}| < N$, where $(\mathbf{O}_{\mathcal{G}}^{\mathbf{M}})_{|\mathcal{F}|}^{-1}$ represents the first $|\mathcal{F}|$ columns of $(\mathbf{O}_{\mathcal{G}}^{\mathbf{M}})^{-1}$, then for any signal $\mathbf{x} \in \mathrm{BL}_{|\mathcal{F}|}(\mathbf{O}_{\mathcal{G}}^{\mathbf{M}})$, the perfect recovery condition is given by [16]

$$\mathbf{x} = \mathbf{R}\mathbf{x}_{\mathcal{S}} = \mathbf{R}\mathbf{D}\mathbf{x}. \tag{3}$$

2.2 Joint Time-Vertex Linear Canonical Transform

The time-vertex signal $\mathbf{X}$ and its vectorized form $\mathbf{x} = \mathrm{vec}(\mathbf{X}) \in \mathbb{C}^{NT}$ represent a graph signal sampled at regular intervals of unit length T [9]. Here, $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T] \in \mathbb{C}^{N\times T}$ is the time-varying graph signal matrix, where $\mathbf{x}_t \in \mathbb{C}^N$ denotes the signal at time t. N is the number of vertices in the graph $\mathcal{G}$, and T is the time series dimension at each vertex. We then use the GLCT on the cyclic graph to define the DLCT, whose parameter matrix is $\mathbf{K}$ and its corresponding parameters $(\bar{\xi}, \bar{\sigma}, \bar{\alpha})$. Thus, the JLCT is defined by the parameter matrices $(\mathbf{K}, \mathbf{M})$ as follows [14]

$$\hat{\mathbf{X}} = \mathrm{JLCT}\{\mathbf{X}\} := \mathbf{O}_{\mathcal{G}}^{\mathbf{M}}\mathbf{X}\left(\mathbf{O}_{\mathcal{D}}^{\mathbf{K}}\right)^{\mathrm{H}}, \tag{4}$$

where $\mathbf{O}_{\mathcal{D}}^{\mathbf{K}}$ is the DLCT operator defined in the time domain and $(\cdot)^{\mathrm{H}}$ denotes the conjugate transpose of the matrix. Moreover, $\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}$ is the GLCT operator defined in Eq. (1).

Similar to the JFT, by vectorizing the signal $\mathbf{X}$ into $\mathbf{x} = \mathrm{vec}(\mathbf{X})$, we can identify that the equivalent transformation between the vectorized forms maps from $\mathbb{C}^{NT}$ to $\mathbb{C}^{NT}$.

Definition 2. *For* $\mathbf{x} \in \mathbb{C}^{NT}$, *the JLCT of* $\mathbf{x}$ *is defined as*

$$\hat{\mathbf{x}} = \mathbf{J}^{\mathbf{K},\mathbf{M}}\mathbf{x} = \left(\mathbf{O}_{\mathcal{D}}^{\mathbf{K}} \otimes \mathbf{O}_{\mathcal{G}}^{\mathbf{M}}\right)\mathbf{x}, \tag{5}$$

where $\left(\mathbf{O}_{\mathcal{D}}^{\mathbf{K}} \otimes \mathbf{O}_{\mathcal{G}}^{\mathbf{M}}\right)\mathbf{x} = vec\{\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}\mathbf{X}\left(\mathbf{O}_{\mathcal{D}}^{\mathbf{K}}\right)^{\mathrm{H}}\}$, *and* $\otimes$ *denotes Kronecker product.*

When $\xi = \bar{\xi} = 0$ and $\sigma = \bar{\sigma} = 1$, the JLCT degenerates to the JFRFT [14]. Further setting $\alpha = \bar{\alpha} = 1$ simplifies it to the JFT [17].

3 Sampling in The JLCT Domain

We propose a joint time-vertex sampling method based on the JLCT, building on the GLCT sampling theory [16]. The JLCT extends traditional sampling theory by introducing a flexible concept of bandlimitedness, enabling unified treatment of signals across time and vertex domains. This method is particularly effective for analyzing complex signals, such as non-stationary and chirp-like signals, which are challenging for traditional Fourier-based methods like JFT [10].

The key contribution of JLCT lies in combining the ability of GLCT ability to model scaling and chirp modulation in the graph domain with DLCT in the time domain. This tensor product structure enables JLCT to capture intricate temporal and spatial correlations, making it well-suited for dynamic systems and multi-scale signals.

3.1 Sampling and Recovery

For the time-vertex signal $\mathbf{X}$, we can consider its vectorized form $\mathbf{x} = \mathrm{vec}(\mathbf{X})$, with sampling represented as

$$\mathbf{x}_{\mathcal{S}_J} = \mathbf{D}_J\mathbf{x}, \tag{6}$$

where $\mathbf{D}_J \in \{0, 1\}^{|\mathcal{S}_J|\times NT}$ is the sampling operator similarly defined as Eq. (2), and $\mathcal{S}_J$ represents the sample set of the time-vertex signal. We can perfectly recover $\mathbf{x}$ from $\mathbf{x}_{\mathcal{S}_J}$, using the recovery operator $\mathbf{R}_J \in \mathbb{C}^{NT\times|\mathcal{S}_J|}$, which is a linear mapping from $\mathbb{C}^{|\mathcal{S}_J|}$ to $\mathbb{C}^{NT}$. Similar to the GLCT sampling [16], we present a theorem for perfect recovery of time-vertex signals in the JLCT domain.

Theorem 1. *For any time-vertex signal* $\mathbf{x}$ *bandwidth-limited to* $|\mathcal{F}_J|$, *perfect recovery is achievable if* $\mathrm{rank}\{\mathbf{D}_J(\mathbf{J}^{\mathbf{K},\mathbf{M}})_{|\mathcal{F}_J|}^{-1}\} = |\mathcal{F}_J| < NT$. *The signal can be reconstructed as*

$$\mathbf{x} = \mathbf{R}_J\mathbf{x}_{\mathcal{S}_J} = (\mathbf{J}^{\mathbf{K},\mathbf{M}})_{|\mathcal{F}_J|}^{-1}\mathbf{P}_J\mathbf{x}_{\mathcal{S}_J}, \tag{7}$$

where $\mathbf{P}_J = \{\mathbf{D}_J(\mathbf{J}^{\mathbf{K},\mathbf{M}})_{|\mathcal{F}_J|}^{-1}\}^{\dagger}$, *and* $(\mathbf{J}^{\mathbf{K},\mathbf{M}})_{|\mathcal{F}_J|}^{-1}$ *corresponds to the first* $|\mathcal{F}_J|$ *columns of* $(\mathbf{J}^{\mathbf{K},\mathbf{M}})^{-1}$.

Proof. The rank condition $\mathrm{rank}\{\mathbf{D}_J(\mathbf{J}^{\mathbf{K},\mathbf{M}})_{|\mathcal{F}_J|}^{-1}\} = |\mathcal{F}_J|$ ensures that the recovery operator $\mathbf{R}_J = (\mathbf{J}^{\mathbf{K},\mathbf{M}})_{|\mathcal{F}_J|}^{-1}\mathbf{P}_J$ spans the subspace $\mathrm{BL}_{|\mathcal{F}_J|}(\mathbf{J}^{\mathbf{K},\mathbf{M}})$. Since $\mathbf{R}_J$ is a projection operator, the signal can be perfectly recovered when it belongs to the corresponding bandlimited subspace. □

For time-vertex signals, the vectorized form $\mathbf{x} \in \mathbb{C}^{NT}$ corresponds to $\mathbf{J}^{\mathbf{K},\mathbf{M}} \in \mathbb{C}^{NT\times NT}$. Given the high dimensionality, frequency analysis and sampling are computationally demanding.

To address this, we propose a novel direct sampling method on $\mathbf{X}$, reducing complexity and enabling efficient handling of high-dimensional signals. Next, we define bandwidth.

Definition 3. *For a generally bandlimited signal* $\mathbf{x} \in \mathbb{C}^{NT}$*: (i) The overall bandwidth is* $|\mathcal{F}_J|$ *when* $\hat{\mathbf{x}}$ *has fewer than* $|\mathcal{F}_J| < NT$ *nonzero elements; (ii) The bandwidth projected onto* $\mathcal{G}_T$ *is* $|\mathcal{F}_T|$ *when* $\hat{\mathbf{X}}$ *exhibits* $|\mathcal{F}_T|$ *nonzero rows; (iii) The bandwidth projected onto* $\mathcal{G}_G$ *is* $|\mathcal{F}_G|$ *when* $\hat{\mathbf{X}}$ *contains* $|\mathcal{F}_G|$ *nonzero columns.*

The projected bandwidth highlights the relationship between $\mathcal{G}_J$, $\mathcal{G}_T$, and $\mathcal{G}_G$. As described in Definition 3, a signal $\mathbf{X}$ is considered synchronously bandlimited if the projected bandwidth satisfies $|\mathcal{F}_T| < T$ and $|\mathcal{F}_G| < N$. Clearly, the general relationship is $\max(|\mathcal{F}_T|, |\mathcal{F}_G|) \le |\mathcal{F}_J| \le |\mathcal{F}_T||\mathcal{F}_G|$.

Building on Theorem 1, and considering

$$\mathrm{rank}(\mathbf{O}^{\mathbf{K}}_{\mathcal{D}})_{|\mathcal{F}_T|} \otimes (\mathbf{O}^{\mathbf{M}}_{\mathcal{G}})_{|\mathcal{F}_G|} = |\mathcal{F}_J| = \mathrm{rank}(O^{\mathbf{K}}_{\mathcal{D}})_{|\mathcal{F}_T|}\mathrm{rank}(\mathbf{O}^{\mathbf{M}}_{\mathcal{G}})_{|\mathcal{F}_G|},$$

let $\mathcal{S}_G \subseteq \mathcal{V}_G$ and $\mathcal{S}_T \subseteq \mathcal{V}_T$ be vertex subsets from $\mathcal{G}_T$ and $\mathcal{G}_G$. There exist suitable sampling sets $|\mathcal{S}_T| \ge |\mathcal{F}_T|$ and $|\mathcal{S}_G| \ge |\mathcal{F}_G|$, enabling the recovery of the signal $\mathbf{X}$ from the sampled signal as

Let $\mathcal{S}_G \subseteq \mathcal{V}_G$ and $\mathcal{S}_T \subseteq \mathcal{V}_T$ be sampling subsets of graph and time vertices, respectively. The sampled signal is:

$$\mathbf{X}_{\mathcal{S}_G \times \mathcal{S}_T} = \mathbf{D}_G \mathbf{X} \mathbf{D}_T^{\mathrm{H}} = \mathbf{D}_G (\mathbf{O}^{\mathbf{M}}_{\mathcal{G}})^{-1}_{|\mathcal{F}_G|} \hat{\mathbf{X}}_{|\mathcal{F}_J|} (\mathbf{O}^{\mathbf{K}}_{\mathcal{D}})_{|\mathcal{F}_T|} \mathbf{D}_T^{\mathrm{H}}, \quad (8)$$

where $\mathbf{D}_G$ and $\mathbf{D}_T$ are the sampling matrices corresponding to the sets $\mathcal{S}_G$ and $\mathcal{S}_T$. Thus, the reconstructed signal is

$$\mathbf{X}_{\mathcal{R}_G \times \mathcal{R}_T} = \mathbf{R}_G \mathbf{X}_{\mathcal{S}_G \times \mathcal{S}_T} \mathbf{R}_T^{\mathrm{H}} = \mathbf{R}_G \mathbf{D}_G \mathbf{X} \mathbf{D}_T^{\mathrm{H}} \mathbf{R}_T^{\mathrm{H}}. \quad (9)$$

This enables perfect recovery of $\mathbf{X}$ from $\mathbf{X}_{\mathcal{S}_G \times \mathcal{S}_T}$ using the recovery operators $\mathbf{R}_G$ and $\mathbf{R}_T$, which map $\mathbb{C}^{|\mathcal{S}_G| \times |\mathcal{S}_T|}$ to $\mathbb{C}^{N \times T}$.

The actual sampling set for $\mathcal{G}_J$ can be expressed as $\mathcal{S}_J = \mathcal{S}_T \times \mathcal{S}_G$, with the total number of samples being $|\mathcal{S}_J| = |\mathcal{S}_T||\mathcal{S}_G|$. The vectorized form of $\mathbf{X}_{\mathcal{S}_G \times \mathcal{S}_T}$, can be expressed as

$$\mathbf{x}_{\mathcal{S}_J} = \left\{ \mathbf{D}_T (\mathbf{O}^{\mathbf{K}}_{\mathcal{D}})^{-1}_{|\mathcal{F}_T|} \otimes \mathbf{D}_G (\mathbf{O}^{\mathbf{M}}_{\mathcal{G}})^{-1}_{|\mathcal{F}_G|} \right\}^{\mathrm{H}} \hat{\mathbf{x}}_{|\mathcal{F}_J|}. \quad (10)$$

Therefore, the sampling matrix for the set $\mathcal{S}_J$ is given by

$$\mathbf{D}_J = \mathbf{D}_T \otimes \mathbf{D}_G, \quad (11)$$

which is consistent with Eq. (6). The corresponding recovery operator $\mathbf{R}_J = \mathbf{R}_T \otimes \mathbf{R}_G$ also aligns with Eq. (7).

3.2 Optimal Sampling Set Selection

To sample time-vertex signals, it is crucial to determine the appropriate sample size and develop strategies for selecting optimal sampling nodes. For the vectorized signal $\mathbf{x}$, this involves choosing $|\mathcal{F}_J|$ linearly independent rows from $(\mathbf{J}^{\mathbf{K},\mathbf{M}})^{-1}_{|\mathcal{F}_J|}$ to minimize noise impact [16]. Sampling is represented as

$$\mathbf{x}_{\mathcal{S}_J} = \mathbf{D}_J \mathbf{x} + e,$$

where e denotes noise. The recovered signal is then

$$\mathbf{x}_{\mathcal{R}} = \mathbf{R}_J \mathbf{x}_{\mathcal{S}_J} = \mathbf{R}_J \mathbf{D}_J \mathbf{x} + \mathbf{R}_J e = \mathbf{x} + \mathbf{R}_J e.$$

By using the original signal x, the recovery error is

$$\begin{aligned} ||\mathbf{x}_{\mathcal{R}} - \mathbf{x}||_2 &= ||\mathbf{R}_J e||_2 \le ||\mathbf{R}_T \otimes \mathbf{R}_G||_2 \, ||e||_2 \\ &= \left\| (\mathbf{O}^{\mathbf{K}}_{\mathcal{D}})^{-1}_{|\mathcal{F}_T|} \mathbf{P}_T \otimes (\mathbf{O}^{\mathbf{M}}_{\mathcal{G}})^{-1}_{|\mathcal{F}_G|} \mathbf{P}_G \right\|_2 ||e||_2 \\ &= \left\| (\mathbf{J}^{\mathbf{K},\mathbf{M}})^{-1}_{|\mathcal{F}_J|} \{\mathbf{D}_J (\mathbf{J}^{\mathbf{K},\mathbf{M}})^{-1}_{|\mathcal{F}_J|}\}^{\dagger} \right\|_2 ||e||_2 \\ &\le \left\| (\mathbf{J}^{\mathbf{K},\mathbf{M}})^{-1}_{|\mathcal{F}_J|} \right\|_2 \left\| \{\mathbf{D}_J (\mathbf{J}^{\mathbf{K},\mathbf{M}})^{-1}_{|\mathcal{F}_J|}\}^{\dagger} \right\|_2 ||e||_2 , \end{aligned}$$

where $\mathbf{P}_G = \{\mathbf{D}_G (\mathbf{O}^{\mathbf{M}}_{\mathcal{G}})^{-1}_{|\mathcal{F}_G|}\}^{\dagger}$ and $\mathbf{P}_T = \{\mathbf{D}_T (\mathbf{O}^{\mathbf{K}}_{\mathcal{D}})^{-1}_{|\mathcal{F}_T|}\}^{\dagger}$. The optimal sampling operator $\mathbf{D}_J$ maximizes the minimum singular value of $\mathbf{D}_J (\mathbf{J}^{\mathbf{K},\mathbf{M}})^{-1}_{|\mathcal{F}_J|}$, ensuring robustness to noise

$$\mathbf{D}_J^{\mathrm{opt}} = \arg\max_{\mathbf{D}_J} \sigma_{\min}(\mathbf{D}_J (\mathbf{J}^{\mathbf{K},\mathbf{M}})^{-1}_{|\mathcal{F}_J|}). \quad (12)$$

This approach ensures that the sampling operator $\mathbf{D}_J$ is optimal for signal recovery. However, as networks and graphs grow more complex and higher-dimensional, computational cost and complexity increase significantly.

To address this, we propose *distributed bandwidth partitioning*, which independently optimizes the sampling sets $\mathbf{D}_G$ and $\mathbf{D}_T$ in the graph and time domains. This method reduces computational complexity while ensuring robust recovery, making it suitable for large matrix signals. For high-dimensional data $\mathbf{x}$, we extend this approach to select the optimal sampling set for $\mathbf{X}$, considering its noise matrix $\mathbf{E}$, as follows

$$\begin{aligned} &||\mathbf{X}_{\mathcal{R}_G \times \mathcal{R}_T} - \mathbf{X}||_2 = ||\mathbf{R}_G \mathbf{E} \mathbf{R}_T^{\mathrm{H}}||_2 \\ &= \left\| (\mathbf{O}^{\mathbf{M}}_{\mathcal{G}})^{-1}_{|\mathcal{F}_G|} \mathbf{P}_G \mathbf{E} \mathbf{P}_T^{\mathrm{H}} (\mathbf{O}^{\mathbf{K}}_{\mathcal{D}})_{|\mathcal{F}_T|} \right\|_2 \\ &\le \left\| (\mathbf{O}^{\mathbf{M}}_{\mathcal{G}})^{-1}_{|\mathcal{F}_G|} \right\|_2 ||\mathbf{P}_G||_2 \, ||\mathbf{E}||_2 \left\| \mathbf{P}_T^{\mathrm{H}} \right\|_2 \left\| (\mathbf{O}^{\mathbf{K}}_{\mathcal{D}})_{|\mathcal{F}_T|} \right\|_2 . \end{aligned}$$

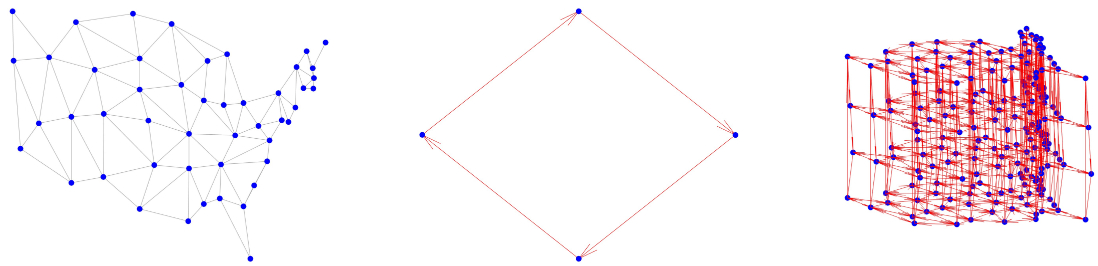
The optimal sampling sets in the graph and time domains minimize $\|\mathbf{P}_G\|_2$ and $\|\mathbf{P}_T^{\mathrm{H}}\|_2$, respectively,

$$(\mathbf{D}_G^{\mathrm{opt}}, \mathbf{D}_T^{\mathrm{opt}}) = \arg\min_{\mathbf{D}_G, \mathbf{D}_T} \|\mathbf{P}_G\|_2 \cdot \|\mathbf{P}_T^{\mathrm{H}}\|_2. \quad (13)$$

The JLCT framework, combined with distributed bandwidth partitioning, effectively handles high-dimensional non-stationary and chirp-like signals with time-varying frequency components. By capturing both temporal and spatial features, JLCT provides a powerful tool for dynamic signal processing, such as time-varying sensor networks and evolving social networks. A greedy algorithm [5, 16] can be used to select optimal sampling nodes, as shown in Algorithms 1 and 2.

Algorithm 1 Optimal Sampling Operator Based on Vector $\mathbf{x}$

Input: $(\mathbf{J}^{\mathbf{K},\mathbf{M}})^{-1}_{|\mathcal{F}_J|}$, $|\mathcal{S}_J|$, N
Output: $\mathcal{S}_J$
while $|\mathcal{S}_J| < |\mathcal{S}_J|$ **do**
  $s = \arg\max_{i \in \{1,\dots,N\} \setminus \mathcal{S}_J} \sigma_{\min}\left(\left[(\mathbf{J}^{\mathbf{K},\mathbf{M}})^{-1}_{|\mathcal{F}_J|} \right]_{\mathcal{S}_J \cup \{i\}} \right)$
  $\mathcal{S}_J = \mathcal{S}_J \cup \{s\}$
end while
return $\mathcal{S}_J$

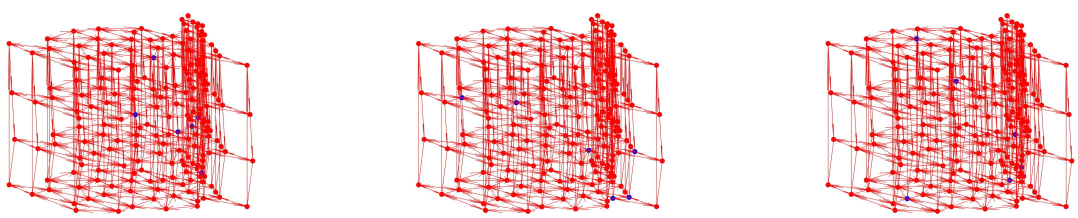


(a) 48-note underlying graph.

(b) 4-note time series cyclic graph.

(c) 192-note product graph.

Figure 1: The joint time-vertex graph.



(a) Samples from JFT sampling with $(\alpha, \sigma, \xi, \bar{\alpha}, \bar{\sigma}, \bar{\xi}) = (1, 1, 0, 1, 1, 0)$.

(b) Samples from JFRFT sampling with $(\alpha, \sigma, \xi, \bar{\alpha}, \bar{\sigma}, \bar{\xi}) = (0.7, 1, 0, 0.4, 1, 0)$.

(c) Samples from JLCT sampling with $(\alpha, \sigma, \xi, \bar{\alpha}, \bar{\sigma}, \bar{\xi}) = (0.7, 1, -0.5, 0.4, 3, 1)$.

Figure 2: Locations of the sampling nodes (blue dots).

Algorithm 2 Optimal Sampling Operator Based on Matrix $\mathbf{X}$

Input: $(\mathbf{O}_{\mathcal{G}}^{\mathbf{M}})^{-1}_{|\mathcal{F}_G|}$, $(\mathbf{O}_{\mathcal{D}}^{\mathbf{K}})^{-1}_{|\mathcal{F}_T|}$, $|\mathcal{S}_G|$, $|\mathcal{S}_T|$, N, T
Output: $\mathcal{S}_G$ and $\mathcal{S}_T$
while $|\mathcal{S}_G| + |\mathcal{S}_T| < |\mathcal{S}_G| + |\mathcal{S}_T|$ **do**
  $s_G = \arg\min_{i \in \{1,\dots,N\} \setminus \mathcal{S}_G} \left\| \left[(\mathbf{O}_{\mathcal{G}}^{\mathbf{M}})^{-1}_{|\mathcal{F}_G|} \right]^{\dagger}_{\mathcal{S}_G \cup \{i\}} \right\|_2$
  $\mathcal{S}_G = \mathcal{S}_G \cup \{s_G\}$
  $s_T = \arg\min_{j \in \{1,\dots,T\} \setminus \mathcal{S}_T} \left\| \left[(\mathbf{O}_{\mathcal{D}}^{\mathbf{K}})^{-1}_{|\mathcal{F}_T|} \right]^{\dagger}_{\mathcal{S}_T \cup \{j\}} \right\|_2$
  $\mathcal{S}_T = \mathcal{S}_T \cup \{s_T\}$
end while
return $\mathcal{S}_G$ and $\mathcal{S}_T$

Then, we perform simulation experiments to demonstrate the effectiveness of our proposed sampling method.

The JLCT excels in processing signals that are challenging for traditional Fourier methods, and its ability to integrate time and graph domain information enhances its efficiency. We demonstrate the effectiveness of the proposed sampling method through simulations.

4 Simulation

In this section, we evaluate the effectiveness of the proposed JLCT-based graph signal sampling scheme through extensive simulations. All experiments were conducted using the GSP toolbox [18]. First, we collected precipitation, sunshine, and humidity data for each season from the website[1] for the U.S. states. Since Alaska and Hawaii are too far from the other states, we model the signals on a graph $\mathcal{G}$ representing the geographical layout of the 48 contiguous U.S. states, where each node corresponds to a state [19, 20]. The graph is constructed by connecting neighboring states with undirected edges of unit weight, representing direct geographical adjacency, as shown in Fig. 1.

In this study, we achieve signal recovery using a small number of sampling nodes. By applying Definition 1, we limit the signal bandwidth within the JFT-based bandlimited range, ensuring the signal validity. Set $|\mathcal{S}_G| = |\mathcal{F}_G| = 6$, and $|\mathcal{S}_T| = |\mathcal{F}_T| = 1$. Based on prior knowledge [14–16], the JLCT parameters were set as $\alpha, \bar{\alpha} \in [0, 1]$, $\sigma, \bar{\sigma} \in [1, 7]$, and $\xi, \bar{\xi} \in [-1, 1]$. Using these parameters, the optimal sampling set was determined via Algorithm 2, and signal recovery was performed based on Theorem 1. The recovery error is evaluated using NMSE. Fig. 2 shows the locations of the sampling points for the precipitation signal under different parameters.

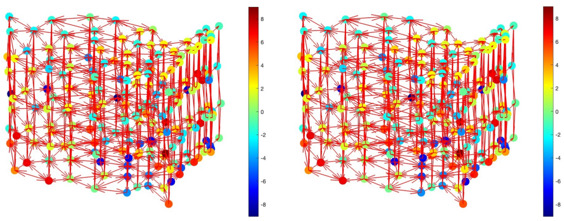
Fig. 3 compares the precipitation signal with the recovery signal using the optimal JLCT parameters. Table 1 presents the optimal JLCT parameters and the minimum recovery error values for different signals. Notably, although the signals are within the JFT bandlimited range, their minimum recovery errors are achieved using the optimal JLCT parameters.

Secondly, we analyze the product graph formed by a Swiss roll graph with $N = 128$ nodes and a cyclic graph with $T = 64$ nodes. The JLCT bandlimiting parameters are set as $(\alpha, \sigma, \xi, \bar{\alpha}, \bar{\sigma}, \bar{\xi}) = (0.5, 1.2, 0.7, 0.3, 1.5, 0.5)$, with $|\mathcal{F}_G| = 20$ and $|\mathcal{F}_T| = 10$. The graph signal is constructed as $\mathbf{X} = (\mathbf{O}_{\mathcal{G}}^{\mathbf{M}})^{-1} \mathbf{1}_{N \times T} (\mathbf{O}_{\mathcal{D}}^{\mathbf{K}})$, where $\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}$ and $\mathbf{O}_{\mathcal{D}}^{\mathbf{K}}$ represent graph-specific operators. To investigate the impact of

[1] https://www.currentresults.com/Weather/US/weather-averages-index.php

Table 1: NMSE vs. Different Signals of Varying Parameters $(\alpha, \sigma, \xi, \bar{\alpha}, \bar{\sigma}, \bar{\xi})$.

Signals	JFT	JFT Error	Optimal JFRFT	JFRFT Error	Optimal JLCT	JLCT Error
Precipitation Signals	$(1, 1, 0, 1, 1, 0)$	2.2661×10^{-13}	$(0.7, 1, 0, 0.4, 1, 0)$	1.8183×10^{-13}	$(0.7, 1, -0.5, 0.4, 3, 1)$	1.4334×10^{-13}
Sunshine Signals	$(1, 1, 0, 1, 1, 0)$	7.6416×10^{-14}	$(0.8, 1, 0, 0.4, 1, 0)$	6.2465×10^{-14}	$(0.8, 2, 0.5, 0.4, 6, 0.5)$	5.1937×10^{-14}
Humidity Signals	$(1, 1, 0, 1, 1, 0)$	1.1341×10^{-13}	$(0.8, 1, 0, 0.5, 1, 0)$	1.1277×10^{-13}	$(0.8, 2, 1, 0.5, 1.5, 1)$	1.0166×10^{-13}



(a) Original precipitation signal. (b) Recovery from JLCT sampling.

Figure 3: Comparison of the original signal and the reconstructed signal.

sampling, the number of sampling points is progressively increased under the optimal JLCT sampling strategy.

As depicted in Fig. 4, a notable trend is observed: increasing the number of spatial sampling points $|\mathcal{S}_G|$ leads to a reduction in recovery accuracy, whereas enhancing the number of temporal sampling points $|\mathcal{S}_T|$ significantly improves the reconstruction performance. This highlights the asymmetric sensitivity of the recovery process to spatial and temporal sampling in product graphs.

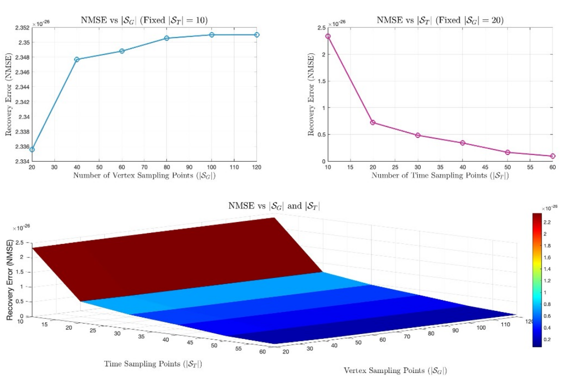


Figure 4: Recovery accuracy vs. Sampling Points.

5 Conclusion

We demonstrate that bandlimited signals in the JLCT domain can be perfectly recovered. Optimal sampling operators on time-vertex graphs are generated using an experimental design strategy. Numerical experiments confirm the superior performance of this approach compared to sampling based on the JFT and the JFRFT. The main results of this paper can be extended to all product graph signals. Future work includes applying JLCT to other dynamic networks, such as time-evolving social or sensor networks, and developing adaptive sampling strategies to enhance efficiency and robustness.

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